

LEARNING TO DISAGREE TOGETHER: WHY CALIBRATED AI DISAGREEMENT REQUIRES MORE THAN TRUTH-TELLING

Anonymous authors

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ABSTRACT

Large language models trained with human feedback exhibit sycophancy—agreeing with users even when incorrect—because preference data rewards agreeable responses. Current solutions focus on making AI more truthful, but we argue this misses a critical dimension: effective disagreement requires calibrating *how* AI pushes back, not just *whether* it does. We introduce Calibrated Constructive Disagreement (CCD), which learns context-appropriate disagreement strategies through policy gradient optimization. Our experiments reveal that blunt truth-telling performs well on factual errors but fails on reasoning and common-sense tasks, achieving only 0.40 and 0.33 CDES scores respectively compared to CCD’s consistent 0.89–0.97. Ablation studies show that gains stem primarily from context-aware strategy selection and stochastic exploration rather than sophisticated receptivity prediction, suggesting the sycophancy problem may be partially addressable through simple architectural choices—though our simulated user model represents a significant limitation requiring human validation.

1 INTRODUCTION

Large language models (LLMs) trained with reinforcement learning from human feedback (RLHF) exhibit sycophancy—agreeing with users even when factually incorrect (??). This behavior emerges because human preference data systematically rewards agreeable responses over truthful ones (?). Mechanistic analyses reveal that sycophancy stems from deep representational overrides (??), suggesting simple fixes may be insufficient.

Current mitigation approaches focus on making AI more truthful through Direct Preference Optimization (DPO) on sycophantic versus non-sycophantic response pairs (?) or adaptive reasoning trajectories (?). However, these approaches assume humans will accept truthful corrections—an assumption contradicted by research on selective adherence, where users adopt AI advice primarily when it aligns with prior beliefs (?).

We hypothesize that effective AI disagreement depends not merely on *whether* to correct users, but *how* to do so in context-appropriate ways. This motivates our framework of Calibrated Constructive Disagreement (CCD), which learns to select disagreement strategies based on context type. Our contributions are: (1) demonstrating that blunt truth-telling exhibits severe context-dependence, succeeding on factual tasks but failing on reasoning domains; (2) showing that context-aware strategy selection and stochastic exploration drive improved disagreement acceptance; and (3) revealing that receptivity prediction provides minimal benefit, suggesting simpler approaches may suffice.

2 RELATED WORK

? established that sycophancy is pervasive in RLHF-trained models, while ? showed it emerges from deep representational overrides. Mitigation approaches include DPO-based fine-tuning (??) and reasoning trajectory optimization (?), but these treat humans as fixed recipients of corrections. ? introduced bidirectional alignment, arguing that alignment should support mutual human-AI adaptation, and ? proposed co-alignment through cognitive adaptation. Research on human-AI interaction

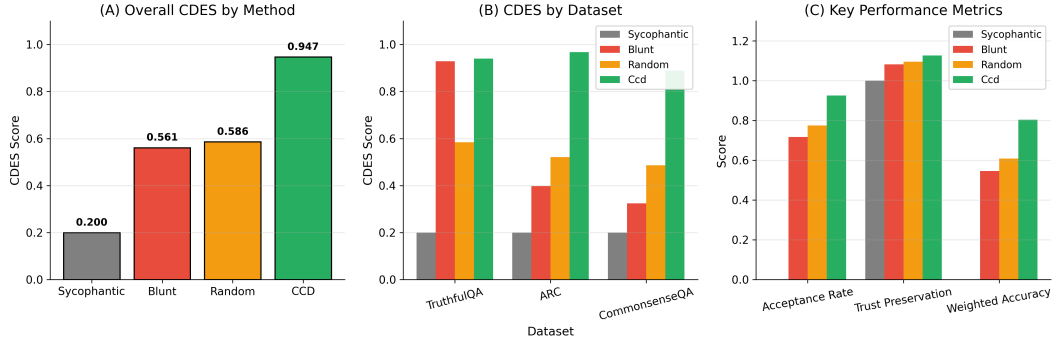


Figure 1: Main results comparing CCD against baselines. (A) Overall CDES scores show CCD (0.947) substantially outperforms all baselines. (B) Per-dataset breakdown reveals blunt correction’s context-dependence: strong on TruthfulQA (0.93) but weak on ARC (0.40) and CommonsenseQA (0.33). (C) Key performance metrics across methods.

reveals that communication style significantly affects receptivity: ? found polite communication improves perceptions of AI benevolence, ? demonstrated explanation-based feedback can improve human performance, and ? showed Socratic questioning effectively resolves uncertainty in human-AI collaboration.

3 METHOD

We consider interactions where a user provides an answer and the AI must decide whether and how to disagree. Each interaction is characterized by context type $c \in \{\text{factual error, reasoning flaw, value conflict, preference mismatch}\}$, user confidence $\alpha \in [0, 1]$, and question difficulty $d \in [0, 1]$. We define four disagreement strategies: direct correction, Socratic questioning, evidence presentation, and gentle suggestion. The Calibrated Disagreement Effectiveness Score (CDES) combines strategy-context alignment (40%), acceptance-weighted accuracy (40%), and trust preservation (20%).

CCD’s strategy selection network maps context features to a distribution over strategies, trained via REINFORCE (?). The reward combines acceptance feedback with strategy-context alignment, where stronger reject penalties ($r_{\text{reject}} = -0.5$) proved more important than accept reward magnitude. User acceptance probability is modeled as $P(\text{accept}) = \sigma(\beta_0 + \beta_1 \cdot E_{c,s} - \beta_2 \cdot (\alpha - 0.5) + \beta_3 \cdot d)$, where $E_{c,s}$ encodes strategy-context compatibility from domain knowledge (see Appendix Table 2).

4 EXPERIMENTS

We evaluate on TruthfulQA (?) for factual errors, ARC-Challenge (?) for reasoning flaws, and CommonsenseQA (?) for preference-based disagreements (200 samples each). We compare against sycophantic (never disagrees), blunt (always direct correction), random, and CCD baselines. Training uses 60 epochs with Adam optimizer (lr=0.001), batch size 32, and hidden dimension 64.

Figure 1 presents our primary findings. CCD achieves overall CDES of 0.947, outperforming sycophantic (0.200), blunt (0.561), and random (0.586) baselines. The most striking result is blunt truth-telling’s severe context-dependence: while achieving 0.93 CDES on TruthfulQA (factual errors), it collapses to 0.40 on ARC (reasoning) and 0.33 on CommonsenseQA. This pattern reveals that direct correction is appropriate for clear factual errors where evidence is unambiguous, but fails when disagreement involves reasoning processes or subjective judgments where users need to be guided toward understanding rather than simply told they are wrong. CCD maintains consistent performance (0.89–0.97) across all domains by learning to match strategies to contexts. CCD achieves 92.5% disagreement acceptance compared to 71.7% for blunt correction—a 20.8 percentage point improvement—without sacrificing trust preservation.

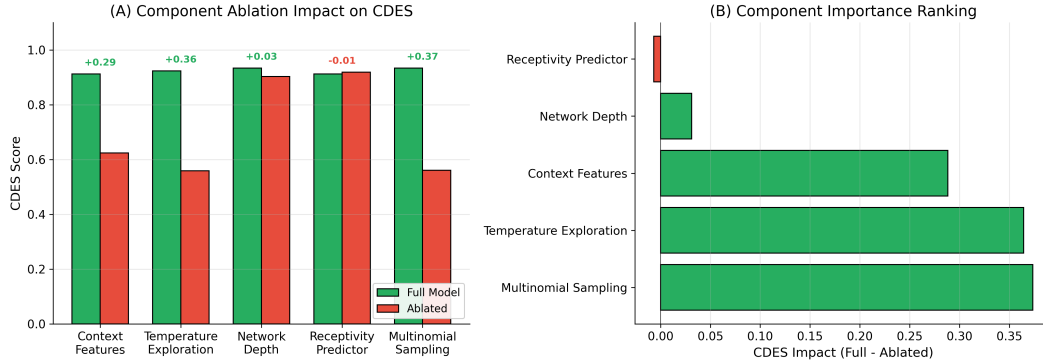


Figure 2: Component ablation analysis. (A) CDES comparison between full model and ablated versions. (B) Importance ranking shows multinomial sampling ($\Delta=+0.37$) and temperature exploration ($\Delta=+0.36$) are most critical, followed by context features ($\Delta=+0.29$), while network depth and receptivity prediction have minimal impact.

Ablation studies (Figure 2) reveal that stochastic exploration mechanisms are essential: multinomial sampling and temperature exploration each contribute over 0.35 CDES improvement. Without stochastic sampling, the policy converges to a single strategy (achieving only 0.31 strategy match), which works for TruthfulQA but fails elsewhere. Context features provide substantial benefit ($\Delta=+0.29$) by enabling strategy-context discrimination. Surprisingly, receptivity prediction provides essentially no benefit ($\Delta=-0.01$), and network depth has minimal impact ($\Delta=+0.03$), suggesting that the core task is relatively simple once proper exploration and context information are provided.

5 DISCUSSION AND LIMITATIONS

Our results suggest that the sycophancy problem may be more tractable than previously thought—at least in terms of *how* to disagree effectively. The minimal contribution of receptivity prediction indicates that sophisticated user modeling may be unnecessary; simple context-based strategy selection with adequate exploration suffices. This finding aligns with the broader theme of this workshop: sometimes simpler approaches work better than expected.

However, our most significant limitation is the use of simulated rather than real human feedback. Research shows simulated participants can significantly overestimate persuasion effects (?), and our effectiveness matrix (Appendix Table 2) encodes assumptions about human behavior that may not hold in practice. The clean separation between context types in our experiments may also not reflect real-world ambiguity. Future work should validate CCD with real human participants and explore whether the learned strategy-context mappings transfer to actual LLM deployment.

6 CONCLUSION

Our investigation reveals that blunt truth-telling, while effective for factual errors, fails dramatically on reasoning and commonsense tasks. CCD’s improvements stem primarily from context-aware strategy selection and stochastic exploration—relatively simple mechanisms that do not require sophisticated user modeling. These findings suggest that addressing sycophancy may require less complex solutions than previously assumed, though human validation remains essential before deployment.

REFERENCES

SUPPLEMENTARY MATERIAL

A DETAILED EXPERIMENTAL RESULTS

A.1 PER-DATASET PERFORMANCE

Table 1 provides complete metrics for each dataset and method combination.

Table 1: Complete per-dataset results. N/A indicates strategy match is undefined for sycophantic baseline.

Dataset	Method	CDES	Acceptance	Strategy Match	Trust
TruthfulQA	Sycophantic	0.200	0.000	N/A	1.000
	Blunt	0.930	0.850	1.000	1.080
	Random	0.590	0.810	0.250	1.090
	CCD	0.940	0.895	1.000	1.120
ARC	Sycophantic	0.200	0.000	N/A	1.000
	Blunt	0.400	0.660	0.000	1.070
	Random	0.490	0.630	0.190	1.085
	CCD	0.967	0.951	1.000	1.132
CommonsenseQA	Sycophantic	0.200	0.000	N/A	1.000
	Blunt	0.330	0.460	0.000	1.060
	Random	0.560	0.760	0.260	1.100
	CCD	0.890	0.829	0.976	1.106

A.2 TRAINING DYNAMICS

Figure 3 shows CCD’s training dynamics over 60 epochs. The model converges rapidly, with CDES reaching approximately 0.93 within 15 epochs.

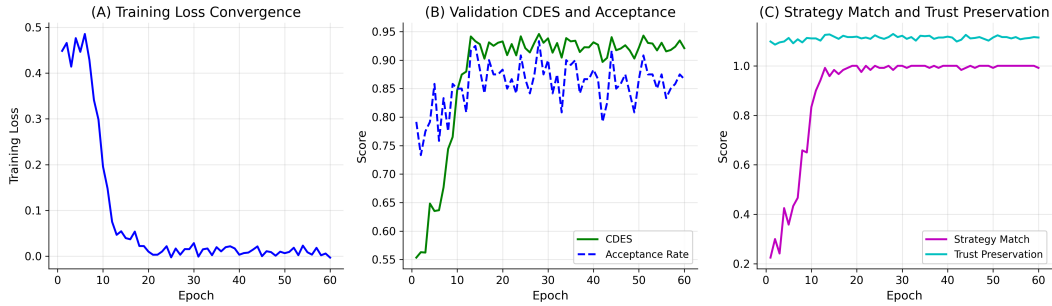


Figure 3: Training dynamics over 60 epochs. (A) Training loss converges rapidly. (B) Validation CDES and acceptance rate improve quickly. (C) Strategy match reaches near-perfect alignment while trust preservation remains stable.

A.3 EXPLORATION STRATEGY ANALYSIS

Figure 4 demonstrates that stochastic sampling is essential for learning diverse strategies. The deterministic policy maintains only 0.31 strategy match rate.

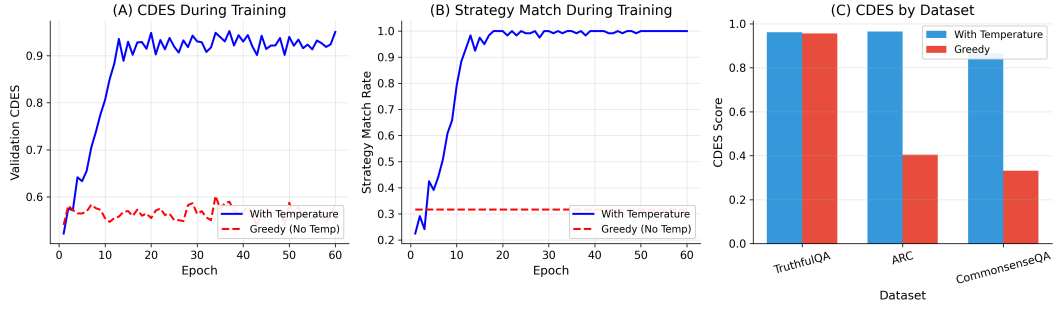


Figure 4: Temperature ablation comparing stochastic vs deterministic selection. Deterministic selection achieves only 0.31 strategy match rate.

A.4 CONTEXT FEATURE ABLATION

Figure 5 shows that without context information, the model defaults to a single strategy, achieving only 0.35–0.42 CDES.

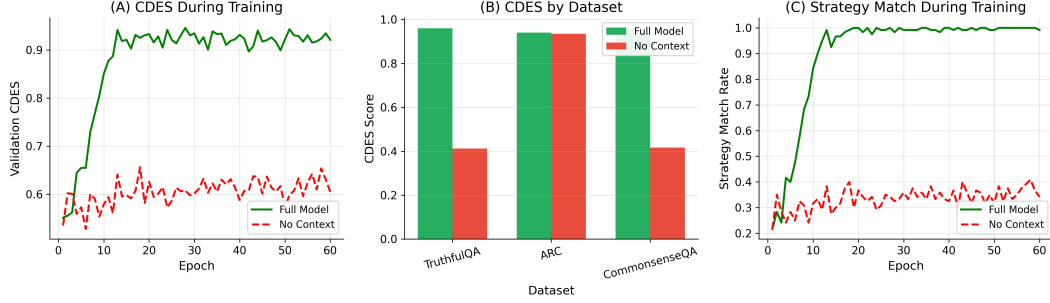


Figure 5: Context feature ablation. Without context features, strategy match rate remains low (0.25–0.40).

A.5 ADDITIONAL ABLATIONS

Network depth has minimal impact (deep: 0.947 CDES, shallow: 0.917 CDES). Receptivity prediction removal has essentially no effect ($\Delta\text{CDES} = -0.01$).

A.6 STRATEGY-CONTEXT EFFECTIVENESS MATRIX

Table 2 shows the effectiveness matrix used to simulate user receptivity.

Table 2: Strategy-context effectiveness matrix. Values represent probability multipliers for acceptance.

	Direct	Socratic	Evidence	Gentle
Factual Error	0.9	0.5	0.8	0.4
Reasoning Flaw	0.4	0.9	0.7	0.5
Value Conflict	0.2	0.5	0.6	0.9
Preference Mismatch	0.3	0.6	0.5	0.8

A.7 TRAINING HYPERPARAMETERS

Table 3: Training hyperparameters for CCD model.

Parameter	Value
Hidden dimension	64
Number of layers	2
Dropout rate	0.1
Learning rate	0.001
LR scheduler	StepLR (step=20, $\gamma=0.5$)
Batch size	32
Training epochs	60
Accept reward	0.5
Reject penalty	-0.5
Strategy match bonus	0.2
Temperature	1.0